



3.5 Billion year old life confirmed

Paleobiologists have confirmed that bacteria and microbes did indeed exist on earth 3.5 Billion years ago. <http://digit.in/oldestbugs>



FDA targets homeopathy

FDA is planning to crackdown on risky homeopathic remedies in the US, especially those sold to treat infants. <http://digit.in/2zki1Pk>

Can genome sequencing bring about truly preventive healthcare?

Arnab Mukherjee | arnab@digit.in

THE GENETIC ANSWER

T

here was a time when doctors mostly had books and their intuition to rely on for diagnosis. Thanks

to the advances of modern medicine, that is no longer the case. Doctors now have an array of tests at their disposal to figure out what's wrong with you. Genetic analysis takes that one step further by letting you figure out which conditions you're vulnerable to before it even occurs.

WHAT IS PERSONAL GENOMICS?

In its application to preventive healthcare in the form of personal genomics, DNA samples are analysed for genes that could indicate the chances of a potentially dangerous condition - using partial or full genome sequencing using gene sequencers. For instance, women with certain variations in the BRCA genes have about an 80 percent chance of developing breast cancer.

ADVANTAGES

"Personal genomics and genomic analysis take advantage of technology that allows the identification of DNA-based risk before the development of signs and symptoms" explains Pooja Ramchandran, Director of Genetic Counselling, Mapmygenome, "For instance, a typical diabetes test will identify elevated levels of blood sugar after it develops and can aid diagnosis. Personal Genomics, on that other hand, facilitates preventive medicine by identifying 'invisible' genetic

risks and allowing the implementation of a plan based on proactive risk reductive strategies of dietary and lifestyle modification"

POPULARITY

Mapmygenome, one of the leaders in personal genomics in India, reports more than 10,000 customers since 2013 for their Genomepatri, which is their most comprehensive offering that provides a complete genome analysis to the individual. Medgenome, another name in the personal genomics segment in India, recently acquired a high-end sequencer. These trends do indicate a growing popularity of genome sequencing in India, although it is still far from being a popular alternative to regular testing. That doesn't mean that efforts aren't being made.

At the CSIR Institute of Genomics and Integrative Biology, Dr Vinod Scaria and his colleague Sridhar Sivasubbu have organised a network of doctors around the country - named GUARDIAN - for the mapping of individuals and families suffering from rare diseases that might have a strong genetic aspect.

REPRODUCTIVE PLANNING

Going one step further, proper support from the government could facilitate extensive genome mapping from around the country. With a large quantity of data available, trends related to medical conditions and genetic diseases can be better observed. Another offering from Mapmygenome,

essentially tests the genes of a potential couple for reproductive planning. "Every individual is a carrier of 1 or 2 serious genetic conditions that can be passed onto the next generation.", says Pooja, "However, if both members of a couple are carriers of the same genetic condition, each pregnancy carries a 1 in 4 (or 25 percent) chance of being affected. Matchmygenome, and genetic carrier screening, helps couples identify their carrier status of around 170 genetic conditions."

In India, which has a high prevalence of genetic conditions including thalassemia and sickle cell disease, a service like Matchmygenome allows would be parents to:

- Determine the chance for future children to be affected by certain hereditary conditions
- Understand the available options for achieving pregnancy
- Easily make informed decisions for the future

THE FUTURE

The next frontier where sequencing technology is to be applied widely is Whole Exome Sequencing (WES). The exome is a subset of the genome that contains the protein-coding region of the human genome. While it represents less than 2 percent of the genome, it contains ~85 percent of known disease-related variants. As a result, WES is more cost, time, and analysis effective as compared to Whole Genome Sequencing and that is where the future lies. 